

Engram and Conditional Memory

Deep Dive into DeepSeek — Chapter 15

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Foundation Books Project

The one rule of this chapter

The *address* is deterministic from the token suffix.

The *contribution* is not.

- Engram stores learned rows *inside the checkpoint*. A compressed token suffix fixes a small set of row addresses.
- The rows are read; then the *current hidden state* gates their contribution before the residual merge.
- That address–read–gate sequence is the object. Only after it is fixed may we classify the path against MoE or retrieval.

Goal: follow one scaled-down lookup-and-gate record from token suffix to residual merge, then attach each of the paper's comparisons to the controls that support it.

Engram is adjacent DeepSeek research, not a component inherited by the released V3–V4 model artifacts.

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2. The address-to-gate path

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3. Collisions, sharing, and deadlines

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4. What the paper's comparisons show
5. Boundaries, code, and what carries forward

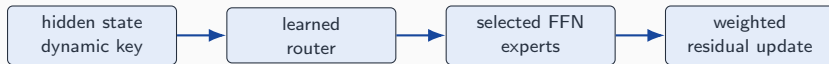
Two axes of sparsity

Three selection mechanisms, not one

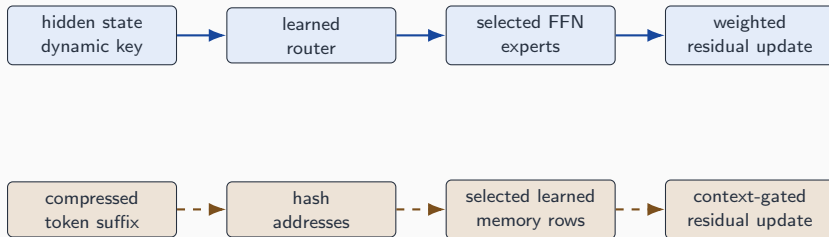
Question	MoE conditional computation	Engram conditional memory	External retrieval / RAG
Selection key	current hidden state scored by a learned router	compressed token suffix mapped by deterministic hashes	query matched against a corpus or search service
Selected object	expert functions F_e	learned embedding rows	documents, chunks, or records
Dynamic decision	expert identities and route weights	context gate applied <i>after</i> deterministic addressing	query construction, ranking, filtering
Update boundary	learned in training or fine-tuning	learned in training or fine-tuning	corpus may change without retraining

The category error to avoid: a large count of inactive parameters describes both an expert bank and a memory table. The expert *executes a selected function*; the lookup row *supplies a retrieved value*.

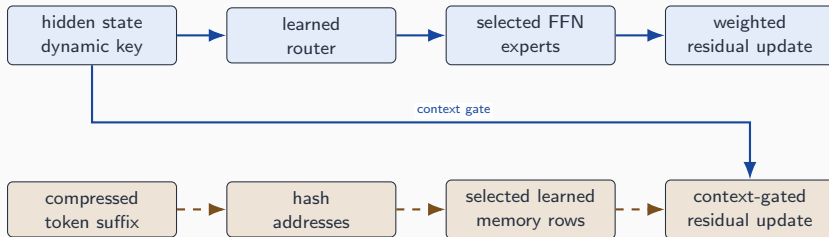
Two dataflows, one picture



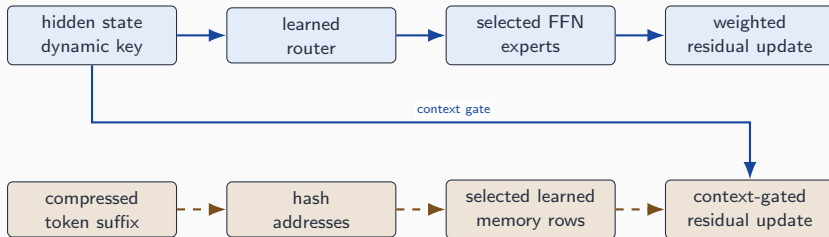
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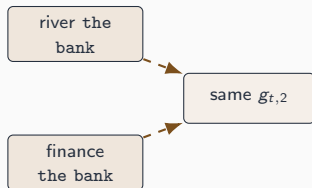


Warning (Predictable address is not unconditional use)

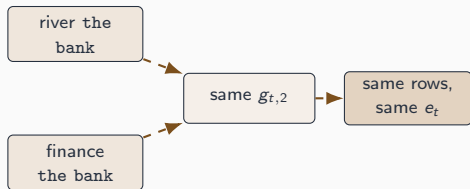
For a fixed token suffix and hash configuration, Engram can determine the row addresses before the target layer executes. The residual contribution still depends on the current hidden state through the context gate. Predictable addressing enables prefetch; the gate preserves conditional use, and row movement still incurs work.

The address-to-gate path

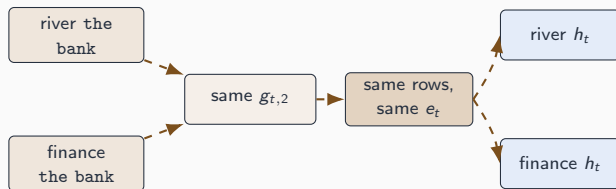
The running record: same suffix, different use



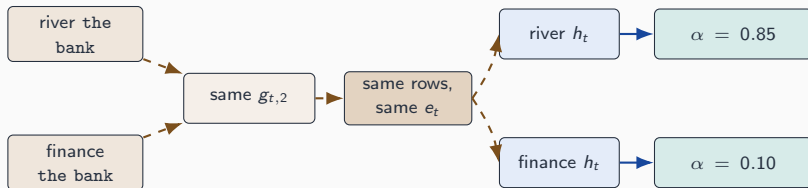
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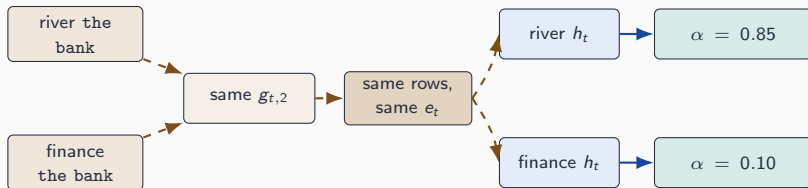
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- Two passages, one river and one finance, both ending in the same compressed bigram "the bank". Fix $n_{\max} = 2$ and the hash configuration.
- Canonicalization, the bigram hash-head addresses, and row selection therefore *agree*. Only the gate differs.

The lookup did not resolve the meaning. The context gate decided how strongly that shared prior entered each residual stream. [DERIVED: scaled-down teaching record; not measured activations]

The deterministic half: suffix, hash, row

With pad-extended ids $\bar{z}_s$ and precomputed canonical ids $C(\bar{z}_s)$,

$$g_{t,n} = (C(\bar{z}_{t-n+1}), \dots, C(\bar{z}_t)), \quad 2 \leq n \leq n_{\max},$$

$$a_{t,n,h} = H_{n,h}(g_{t,n}) \bmod M_{n,h}, \quad e_t = \text{Concat}_{n,h} E_{n,h}[a_{t,n,h}].$$

- Left-padding at early positions keeps the hash-head count and the concatenated projection width fixed.

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[REPORTED: Engram paper]

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Multiple hash heads reduce dependence on any one collision, but collisions remain possible — so the retrieved vector stays a context-independent *prior*, not a trusted fact.

The dynamic half: gate, convolution, residual

$$k_t = W_K e_t, \quad v_t = W_V e_t, \quad \alpha_t = \sigma \left(\frac{\text{RMSNorm}(h_t)^T \text{RMSNorm}(k_t)}{\sqrt{d}} \right), \quad \tilde{v}_t = \alpha_t v_t.$$

$$Y = \text{SiLU} \left(\text{Conv1D} \left(\text{RMSNorm}(\tilde{V}) \right) \right) + \tilde{V}.$$

- In the M -branch integration the table and W_V stay *shared*; each branch owns $W_K^{(m)}$, hence its own gate. Default $M = 4$, with the shared value and four key projections fused into one dense FP8 matrix multiplication. [REPORTED: Engram paper]

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One shared retrieval, M different contextual decisions.

Collisions, sharing, and deadlines

One collision does not alias the record

Hash head	suffix A	suffix B
$H_{2,1}$	5	5
$H_{2,2}$	4	2
$H_{3,1}$	7	0
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- One bigram head aliases the two suffixes; the remaining three heads separate them. The shared first coordinate records the collision, and the complete retrieved priors still differ.
- The scalar gate can suppress a poor retrieval — but it supplies no document provenance, recency, or auditable factual source.

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Multi-head hashing prevents a complete alias; the gate controls use; prefetch changes availability rather than semantics. [DERIVED: toy multiplicative-XOR heads, worked in the chapter]

The prefetch deadline

Let U_{miss} count unique uncached rows after coalescing repeated addresses. For row width d_{head} and s_{dtype} bytes per element, no stall requires

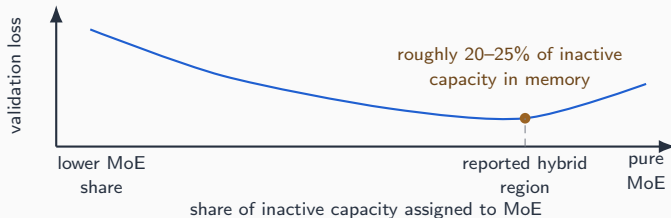
$$T_{\text{hash}} + L_{\text{PCIe}} + \frac{U_{\text{miss}} d_{\text{head}} s_{\text{dtype}}}{BW_{\text{eff}}} \leq \sum_{j < \ell_{\text{Engram}}} T_{\text{block},j}.$$

- Address construction and host transfer may precede the target hidden state. The gate and the residual update may not.
- Deterministic addressing *enables* the overlap. It does not guarantee that a particular cache, interconnect, or workload satisfies the deadline.

Lookup latency hidden inside the independent compute window is overlapped; any remainder extends the critical path.

**What the paper's comparisons
show**

Allocation: a U-shaped trade-off under fixed budgets



- Total- and activated-parameter budgets held fixed; P_{tot} excludes the vocabulary embedding and LM head. Two regimes: 5.7B total/568M active and 9.9B total/993M active, both near a 10:1 sparsity ratio. [REPORTED: Engram paper]
- In the roughly 10B regime, validation loss is 1.7248 for pure MoE and 1.7109 near the 80% MoE / 20% Engram allocation.

Search the trade-off under your own model, data, hardware, and budget — do not copy the reported 20–25% share.

The matched 27B rows

Reported row	MoE-27B	Engram-27B	Difference
Total parameters	26.7B	26.7B	matched
Activated parameters (excl. token emb.)	3.8B	3.8B	matched
Training tokens	262B	262B	matched
Routed experts	72	55	-17
Shared / active routed experts	2 / top-6	2 / top-6	matched
Pile loss	1.960	1.950	-0.010
MMLU, 5-shot accuracy	57.4	60.4	+3.0
BBH, 3-shot exact match	50.9	55.9	+5.0
HumanEval, 0-shot Pass@1	37.8	40.8	+3.0
MATH, 4-shot exact match	28.3	30.7	+2.4
WinoGrande, 5-shot accuracy	67.6	67.8	+0.2

[REPORTED: Engram paper, selected exact rows of its Table 1]

A uniform “improves” summary hides the range: +5.0 on BBH, +0.2 on WinoGrande. Component attribution needs an ablation; deployment performance needs a deployment measurement.

Mechanism, placement, and host residence

- **Effective depth** names a *workload-allocation* hypothesis: an early gated prior may leave more of the unchanged backbone for dynamic composition. Physical depth and attention-head ownership stay fixed.
- **Probes.** LogitLens and CKA support earlier prediction readiness: in the Few-NERD probe, Engram-27B layer 5 aligns most closely with roughly MoE-27B layer 12. [REPORTED: Engram paper]
- **Placement.** 12-layer 3B MoE, 100B tokens, 1.6B Engram budget: single-module insertion favors layer 2 at loss 1.770; a two-module layers-2-and-6 reference reaches 1.768, versus 1.808 for the MoE baseline.
- **Host offload.** A 100B-parameter table in host DRAM, one memory layer in block 2, 512 sequences of length 100–1,024 on H800: 9,031.62 → 8,858.28 tok/s (4B backbone, 1.9%) and 6,315.52 → 6,140.02 tok/s (8B, 2.8%). [REPORTED: Engram paper]
- **Stress test.** Disabling memory after training retains 29–44% on factual tasks and 81–93% on reading comprehension — a train–inference mismatch, not a clean retraining ablation.

Each comparison design supplies only its own controls.

Boundaries, code, and what carries forward

From concept to code: an illustrative map, not a product

The official demonstration is pinned at

from `Engram/engram_demo_v1.py`

https://github.com/deepseek-ai/Engram/blob/fb7f84a21f91223715394a33a1dc24bbfb7f788e/engram_demo_v1.py

- **Addressing.** Canonical token compression and multi-head hashing map to the demonstration's row-lookup path.
- **Contextual use.** The current hidden state maps to the contextual gate; the gated value maps to the short convolution.
- **Boundary.** The README and demonstration module explicitly describe the code as a non-production example that *mocks* the surrounding Attention, MoE, and mHC components.

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So the file clarifies the lookup-and-gate dataflow. It is not a production-readiness claim — that requires separate evidence. Where the revised paper and the demo defaults disagree, the paper is the authority. [INSPECTED: Engram repository, commit fb7f84a]

Concept-to-code map for the chapter

- **lookup, gate, convolution dataflow (mocked surroundings)** → `Engram/engram_demo_v1.py`
- **pad convention and non-production boundary** → `Engram/README.md`
- **architecture, allocation, ablation, offload numbers** → Engram paper, [arXiv:2601.07372v2](#)
- **exact address arithmetic, gate state, kernel tests** → TileKernels chapter, Volume II

Hash to a row; let the hidden state decide how much of it counts.

Next: Volume II's TileKernels chapter takes over the exact prefix-XOR address arithmetic, saved gate state, gradients, and kernel-test surfaces — while the V4 chapter keeps this adjacent research separate from the released V4-Pro and V4-Flash architectures.